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VLSI Physical Design and Power Challenges

Akash Gupta
School of VLSI Design and Embedded System
NIT Kurukshetra
Haryana, India
akashguptakne44@gmail.com

Dr. Chhagan Charan
Department of ECE
NIT Kurukshetra
Haryana, India
chhagan.charan@nitkkr.ac.in

***Abstract*—VLSI is an engineering field that deals with integrated circuit design, verification, and physical implementation. Since the first integrated circuit was created, it has been expanding. Device dimensions might decrease as technology develops. As a result, it became feasible to construct an increasing number of devices on a single integrated circuit. Research on IC has improved performance because of its requirements in computing and communication devices. These days, ICs are measured in nanometers, and work is being done to create better devices. The performance of IC is limited by this compromise. Application-specific integrated chips, or ASICs, are created and optimized for a certain technological node and architectural specifications.**

Low power consumption has become a crucial focus in today's electronics industry. In VLSI chip design, managing power dissipation is now as important as optimizing performance and area. With the rapid growth of battery-powered devices, power dissipation is becoming a major concern in VLSI circuit design. The fast progress in portable and high-performance electronic devices has made it essential to develop low-power VLSI design methods to improve energy efficiency and prolong battery life. In high-performance systems, the power consumed due to leakage is comparable to that used for switching. As chips become smaller and more complex, designing systems that consume less power become more challenging. In low-power VLSI designs, managing leakage current is vital for effective power management. Reducing both leakage and dynamic power is now a key objective in VLSI circuit design.

Keywords—VLSI, gating, cts, routing, scaling, leakage

I. INTRODUCTION

VLSI (Very Large Scale Integration) physical design is an essential phase in the development of semiconductor chips, which serve as the core of electronic devices. This phase involves transforming a circuit's logical blueprint into a manufacturable physical layout. The objective is to efficiently and effectively organize millions, or even billions, of tiny elements like transistors and wires on a chip. The process starts with partitioning, where the circuit is broken down into smaller, more manageable parts. This is followed by floorplanning, akin to arranging furniture in a room, where each part is strategically placed on the chip to optimize space and performance. Next is placement, where individual components are arranged within each section to minimize the distance signals must travel, thereby enhancing speed and reducing power consumption.

Routing then links these components with wires, ensuring seamless signal movement without interference. After placement and routing, the design is verified to ensure it operates correctly and meets all required specifications, a crucial step to identify any errors before production. Throughout the process, various optimization strategies are employed to boost performance, lower power usage, and utilize space efficiently.

Minimizing power consumption is a critical and challenging concern for engineers, scientists, and researchers. In VLSI-CMOS circuit design, power usage is a major issue, with CMOS being the leading technology. The emphasis on low power is driven by the increasing demand for energy-efficient circuit designs in modern devices. However, there is no universal solution that balances power consumption, delay, and area. Consequently, designers must choose appropriate and effective techniques that meet the specific needs of the application and product.

II. SYNTHESIS + FLOORPLAN + PLACEMENT

In the Fusion Complier unified RTL to GDSII we have different steps:

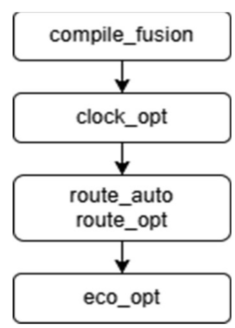

Fig. 1. RTL2GDSII flow in FC

In the complie_fusion step we do the RTL synthesis, floorplan and placement. Different tasks are performed during compile_fusion such as:

- Mapping and area-based optimization followed by the logical based delay and optimization.
- Placement and buffer tree creation.
- Physical optimization for timing, power and area

A. Stages of Compile Fusion

During the compile fusion we have different stages involved:

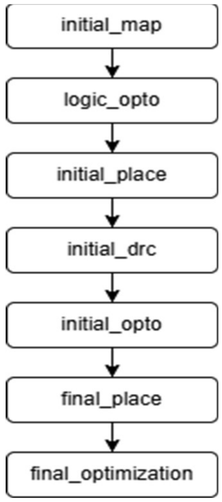

Fig. 2. Compile fusion flow

1) Initial Map:

- It maps the generic logic representation to target library cells and perform area optimization.
- It also inserts the ICG (Integrated Clock Gating) and MV (Multi Voltage) cells.

2) Logic Opto:

- It performs the logical delay optimization and wire length driven placement followed by the more area optimization.
- It also performs the auto-floorplan if needed.

3) Initial Place:

- It performs the buffering aware timing driven placement.
- During this stage, the congestion is not considered.

4) Initial DRC:

- It builds the high fanout buffer trees.
- It includes the buffering of low fanout nets with large DRC violation.

5) Initial Opto:

- It performs the optimization of the placed and HFN inserted netlist.
- It also performs incremental placement as well as incremental HFN buffering and DRC fixing.
- Again, another round of optimization is performed.
- It also enables the CCD (Concurrent Clock Data) optimization.
- It performs the inserted ICG optimization to meet the constrains.
- After all this tasks, end result will be optimised netlist and placed but not fully legalized design database.

6) Final Place:

- It performs the final, incremental timing and congestion driven placement.
- During this stage we can also run the trail CTS.

7) Final Optimization:

- It performs the several rounds of optimization on all the parameters such as timing, power, logical DRC and the CCD optimization.

After the completion of the compile_fusion stages end result will be fully and legalized design which is ready for the CTS.

III. Clock Tree Synthesis (CTS)

Clock Tree Synthesis (CTS) is the process of constructing the clock networks in the design based on the latency/skew specification. In addition to the latency/skew spec. the transition constraints are also used during optimization.

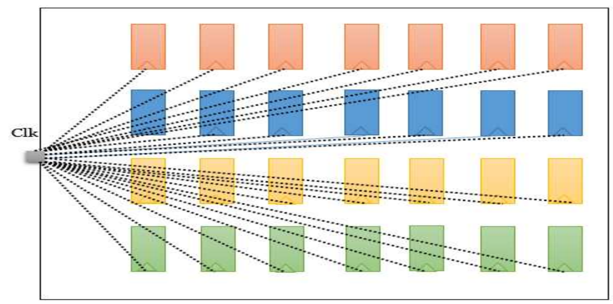

Fig. 3. Before CTS

In this, clock trees are built by taking the considering of the global-skew, local-skew of the clock network and high fanout nets (HFN). The primarily objectives of the clock tree synthesis (CTS) is to reduce the clock skew and clock insertion delays as much as possible. Once the clock trees are build, the routing process of the clock nets will done with any Non Default Rules (NDR) if required.

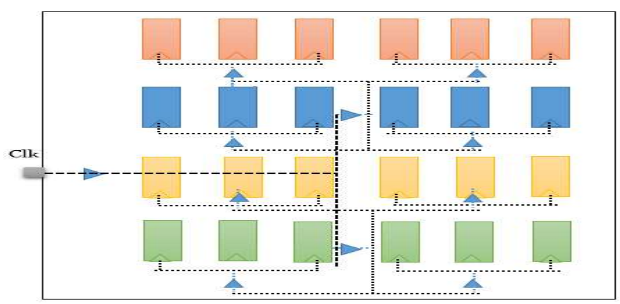

Fig. 4. After CTS

In this, the metal layers to be used and the spacing and width of the metal layers can be specified for the clock network separately. This NDR helps in minimizing the cross-talk of the clock nets since those are the main crucial nets in the design, which are switching more.

At this stage, as the clock trees network are build and actual values of the latency and transition values are obtained. Based on design constraint violations, such as setup and hold violations, the clock tree network is optimized. After the clock tree network is set up, the design is checked for the following violations:

- Setup/hold violations.
- Maximum capacitances violation.
- Maximum transitions violation.
- Maximum fanout violation.
- Signal integrity and cross talk.
- Routing congestion

A. *CTS Structure*

A clock tree is structured to reduce clock skew and make sure the clock signal arrives at all sequential components, like flip-flops and registers, almost simultaneously. Here are some important features of CTS structures in VLSI design.

1) *Fishbone*

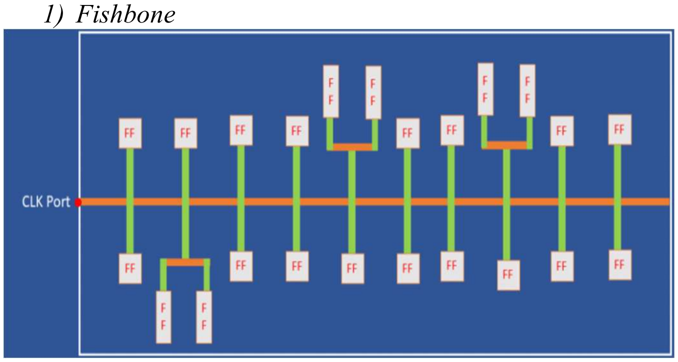

Fig. 5. Fishbone structure of CTS

2) *H – Tree*

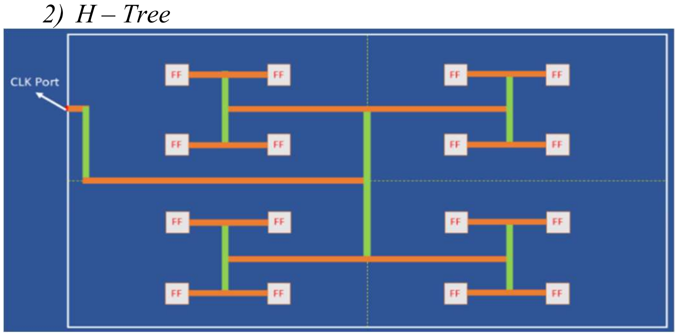

Fig. 6. H-tree

3) *X-Tree*

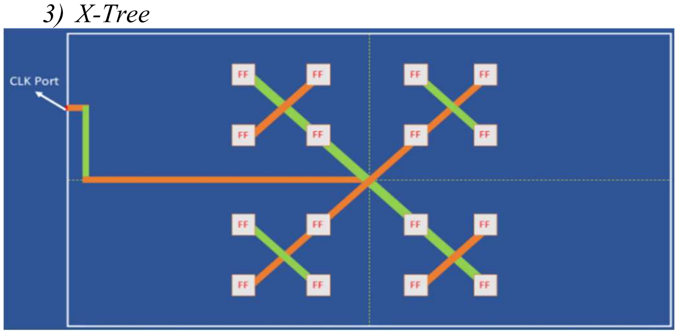

Fig. 7. X-tree

4) *Multi-Source Clock Tree(MSCTS)*

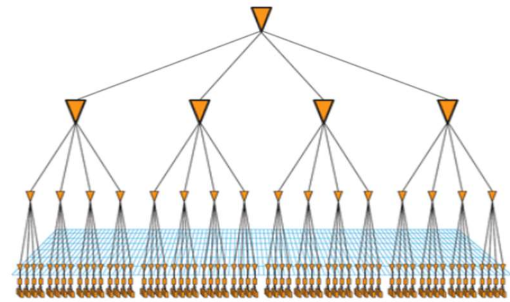

Fig. 8. Multi-source structure of CTS

IV. ROUTING

Routing involves establishing physical connections between signal pins while adhering to design rule checks (DRCs) and considering timing requirements, such as setup and hold time.

A. *Goals of Routing*

- The total interconnect or wire length and the number of vias should be minimized.
- Routing should be completed within the design's specified area.
- There should be no violations of design rule checks (DRCs).
- Timing requirements should be met, and there should be no signoff violation.

B. *Tasks Performed by Routing*

1) *Global Routing*

- It specifies the routing areas.
- It creates an initial path for each net in the design.
- Each net is allocated to a group of routing areas.
- The routing area is divided into tiles or rectangles, known as global routing cells or gcells.
- Global routing allocates nets to specific gcells but does not specify exact tracks within each gcell.
- Typically, the width of the gcells matches the height of a standard cell and aligns with the rows of standard cells.
- Each gcell contains a limited number of horizontal and vertical tracks.
- It does not display the actual wire layout.
- The goal of global routing is to decrease the total wire length and minimize overall overflow.

2) *Track Assignment*

- During the track assignment phase, routing tracks are allocated for each and every global route.
- Tracks are designated in both vertical as well as horizontal orientations.

- Number of available routing layers is determined by the design, and a larger die size allows for more routing tracks.
- Track assignment involves the replacing of all global routes with the specified metal layers.
- This process can lead to numerous violations related to design rules, signal integrity as well as timing.

3) Detailed Routing

- In this phase, the actual connections between/among all the nets are established, including the creation of vias and connections.
- It designates particular tracks for interconnections, with each layers will having its own routing grid and rules.
- Any violations that arose during the track assignment phase are resolved in this stage.
- The primary aim of detailed routing is to finalize all necessary interconnections without any opens, shorts, or spacing violations.
- Detailed routing seeks to minimize the total areas and wire lengths while meeting timing constraints.
- The precise layout of the wires is defined.

C. Route Opt

It performs optimization using VMF (Virtual Metal fill) on routed design for timing, DRC and power and also for hold fixing and CCD optimization. It also adds tap cells around the AON cells. It removes the extraneous pins, ports and terminal.

V. Fill

Chemical Mechanical Polishing (CMP) is one phase in the VLSI manufacturing process that may have an impact on the chip's performance and yield. To ensure that the density of metals and vias should be consistent throughout the scheme, the fill flow incorporates extra metal and via features. Because there is less fluctuation in the polishing depth when the metals and vias are evenly distributed throughout the layout, the negative impacts of the chemical mechanical polishing stages are reduced. Since the fill process includes addition of metals and vias features to the finished layout, this could be affect crucial layout metrics such as timing.

It is the process of adding filler cells and extra metal/vias on empty tracks in the post route db in order to make it DRC and density clean for tape-in and manufacturing. It adds extra metal fill segments to a semiconductor layout. These added shapes are not part of the functional circuitry but are used for uniformity and improve the manufacturing process.

VI. Sign Off

Signoff is the process of verifying the design at final stage before to the tape out. Clean signoff report are the indication for the fabrication because clean signoff report ensure the design satisfies all the required specification and constraints at the final stages.

Signoff tools are only the analysis tools, but PnR tools are the implementation tools. So, all the changes noticed in the signoff stages have to implement only in the PnR tools.

A. Static Timing Analysis (STA)

STA means Static Timing Analysis. STA main function is to make sure that the signal propagates through the design within the time constraints. STA check the design circuit into 4 sets of timing paths, which are reg2reg, in2reg, reg2out and in2out. It then analyze the delay of signal paths in the circuits. By considering these delays, the analysis determines the worst-case and best-case timing for different paths. This is important to make sure that the design is functioning properly and meets all the performance requirements (clock frequency, hold time, setup time).

B. Electromigration (EM) Analysis

It refers to the phenomenon where the movement of metal atoms within a conductor is introduced between the flow of the high current densities. This cause the metal layer to break and can cause shorts. This leads to circuit malfunction or complete failure.

C. Physical Verification

Physical Verification ensure the accurateness of the layout before the manufacturing process. During this stage, tools analyze the design layout data against the predefined design rules which include the metal spacing, metal density, LVS, antenna and many more. By performing this stage, designers can identify the issues early in the design before the tapeout.

D. Formal Equivalance Verification (FEV)

It is a process that compares two versions of a design, such as RTL (Register Transfer Level) and netlist, to confirm they are showing the same behaviour. This process is a subset of formal verification, a broader field that employs mathematical modeling techniques. FEV identifies key points in both models, including primary I/Os, and state elements, and assumes these points should receive the identical values. It then checks that the logical cones which is driving the internal and output crucial points are equal in both the models. FEV is particularly useful for ensuring that a design's functionality remains unchanged after processes like synthesis. In VLSI physical design flow, FEV is employed to verify the design's integrity and correctness before proceeding to manufacturing.

E. Verification Compiler-Low Power (VCLP)

In VLSI, VCLP is a low-power static rule checker designed to validate the implementation of the IEEE 1801 Unified Power Format (UPF) Low-Power design intent. This Synopsys tool ensures that the low-power design intent are accurately implemented and should function as intended. VCLP checks various low-power management techniques and addresses UPF-related challenges, including the management of power domain activities, control signals, supply rail availability, and data retention for functional correctness in the design.

F. Prime Power (PP)

Prime Power evaluates power consumption throughout various design stages, from RTL to the final PnR netlist. During implementation and signoff, Prime Power provides accurate gate-level power analysis reports, helping SoC designers make timely optimizations to meet power goals. It offers both vector-free and vector-based analysis for peak and average power in RTL and gate-level designs. Prime Power provides the complete details on power consumption at the chip, block and cell levels. Average power, peak power, glitched power, clock network power, dynamic and leaky power and multi voltage power are the among the power analysis features.

VII. Low Power In Vlsi

Reducing the power consumption is a crucial and more challenging task for engineers, scientists, and researchers. In VLSI-CMOS circuit design, managing power use is a major concern, with CMOS being the leading technology. The emphasis on low power is driven by the increasing demand for energy-efficient circuit designs in modern devices. However, there is no one-size-fits-all solution that balances power use, delay, and area. As a result, designers must choose effective techniques that meet the specific needs of their applications and products.

Since the invention of the transistor, the microelectronics industry has experienced significant growth, paving the way for devices that use less power. Integrated circuits (ICs) have enhanced circuit performance and reduced their size, leading to increased power density in ICs. The rise of battery-powered, complex medical devices like pacemakers has increased the demand for low-power devices. Researchers are now more focused than ever on developing low-power components and design techniques. Additionally, in high-power devices, the failure rate of silicon doubles with every 10°C increase in temperature. Reducing power consumption has become crucial, especially with the advancement of deep submicron and nanometer technologies. Today, the focus is on low-power devices due to the growth of mobile technology, but even before mobile devices, managing power dissipation was a persistent challenge. High power usage leads to lower performance and reduced reliability of circuits.

A. POWER DISSIPATION

Power dissipations in Complementary Metal Oxide Semiconductor (CMOS) digital circuits can be divided into two categories: peak power and the average power consumption over time. Peak power is a concern for reliability, as it impacts both the lifespan and performance of the chip. When too much instantaneous current flows through the resistive power network, it causes voltage drops that can slow down the design by increasing delays in gates and interconnections. High power usage can also lead to overheating, which reduces the circuit's reliability and lifespan. Additionally, it lowers noise margins, raising the risk of chip failure due to crosstalk.

Average power consumption in traditionally CMOS circuits comes in two forms: dynamic and static. Dynamic power is used when logic gates switch states, as the transistors' internal and external capacitance must be charged, consuming power. Static power is related to gates that are not actively switching. Static power is more significant during standby, particularly for battery-powered devices, whereas dynamic power is crucial during the normal operation, particularly at high frequencies.

$$\boxed{P(\text{total}) = P(\text{dynamic}) + P(\text{short-circuit}) + P(\text{static})}$$

1) Dynamic Power Dissipation

Dynamic power, which primarily comes from charging and discharging parasitic capacitances, consists of three parts: switching power, short-circuit power, and glitching power. Most dynamic power is used when capacitors are charged or discharged. This type of power dissipation does not depend on the size of the transistors but rather on design factors like node capacitance (CL), supply voltage (Vdd), and switching frequency (f).

$$P(\text{dynamic}) = \alpha C_L V_{dd}^2 f$$

Here, C_L is the load capacitance influenced by fanout, wire length, and transistor size. V_{dd} is the supply voltage, which has been decreasing with each new process node, while f is the clock frequency, which has been increasing with each new process node.

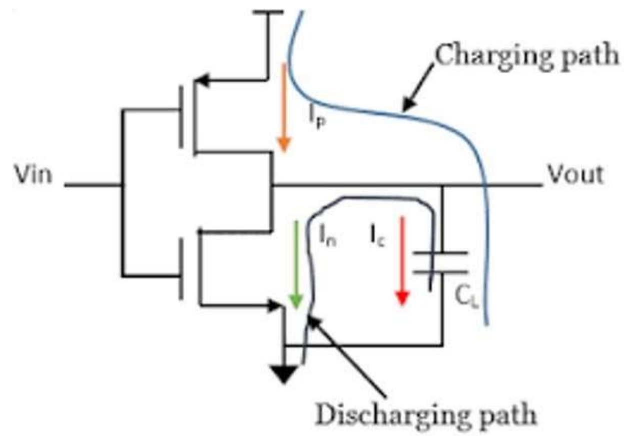

Fig. 9. Dynamic power dissipation

In practice, reducing the supply voltage (V_{dd}) is the most effective power optimization method because switching power has a quadratic relationship with supply voltage, leading to significant power savings. However, lowering the supply voltage can also slow down the circuit's operating speed.

2) Short-Circuit Power Dissipation

When the input changes slowly, there is a period during which some transistors in both the pull-up and pull-down networks are simultaneously turned 'ON,' creating a short-circuit path from VDD to GND.

Short circuit current happens in a CMOS gate during signal transitions when both the nMOS and pMOS networks are active, creating a direct path between Vdd and ground.

- It is directly related to the frequency of switching.
- As the clock frequency increases, frequency of transition/switching of clock increases and the short circuit power dissipation also increases.

- Short-circuit power dissipation is directly proportional to the switching of clock. It is for the very short amount of time.

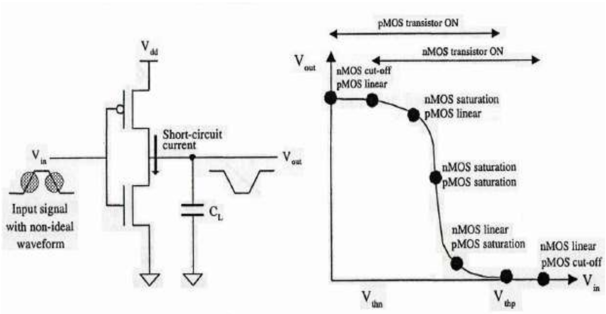

Fig. 10. Short-circuit power dissipation

3) Static power Dissipation

Static power arises from leakage currents when gates are not switching, meaning there are no output changes. Static power dissipation is also known as leakage power. Ideally, CMOS gates ideally shouldn't consume power in this state because either the pull-down or pull-up networks are turned off, stopping static power usage. However, in reality, some leakage current still flows through the transistors, so CMOS gates do use a small amount of power.

While the static power consumed by one logic gate is minimal, it becomes substantial when millions of gates are used in today's integrated circuits (ICs). Additionally, as the transistors are becoming smaller and smaller with each new technology generation, the level of doping increases, leading to higher leakage currents. These leakage currents originate from various sources inside the transistor itself. In long channel transistors, the main contributors of leakage current are reverse diode leakage and subthreshold leakage.

VIII. CONCLUSION

In VLSI (Very Large Scale Integration) design, making efficient and reliable semiconductor chips involves many complex steps, with physical design being a key part. A major challenge in this process is controlling power consumption, particularly as devices become smaller and more powerful.

Designing for low power is crucial because of the growing need for energy-efficient devices, especially in portable electronics. Designers must balance power, performance, and size while ensuring the chip still works properly. Power gating, clock gating, and dynamic voltage scaling are some of the methods used to reduce the power usage. However, these methods need to be applied carefully to ensure they do not impact the chip's speed and reliability.

As technology progresses, finding low power solutions becomes increasingly important, as smaller transistors can cause more leakage currents. Tackling these issues demands creative design approaches and a strong grasp of both the physical and electrical aspects of chip design. The ultimate aim is to develop chips that are not only powerful and compact but also energy-efficient and dependable.

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